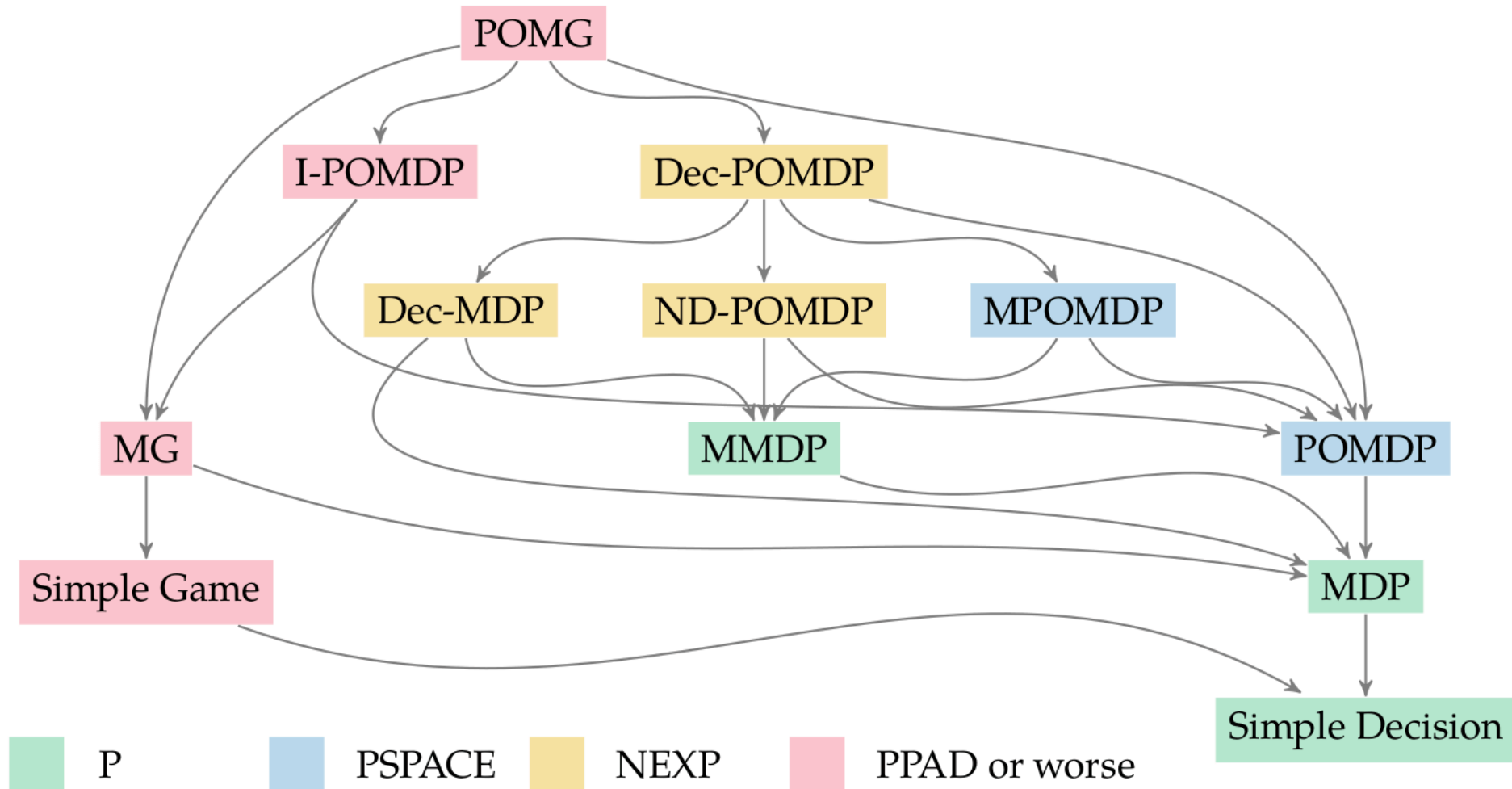


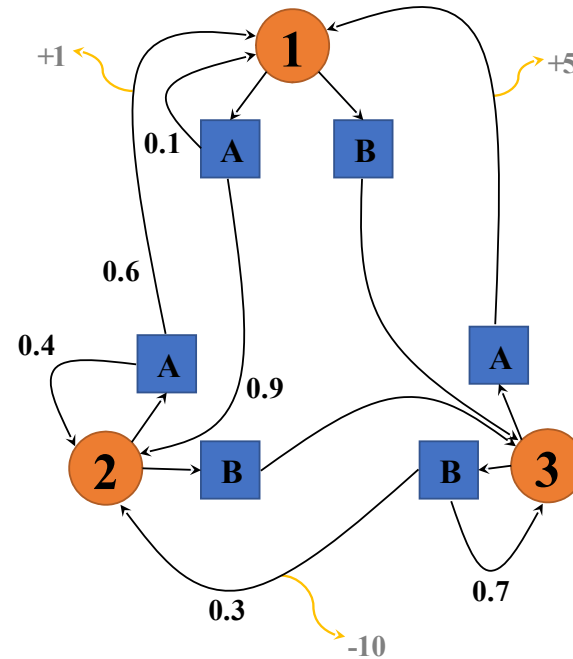
Imperfect Information Dynamic Games

Tyler Becker

Hierarchy of Problems



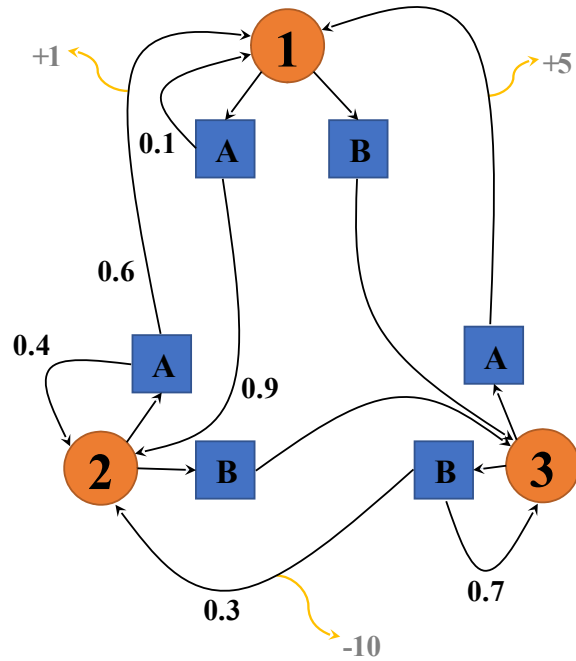
Markov Decision Process (MDP)



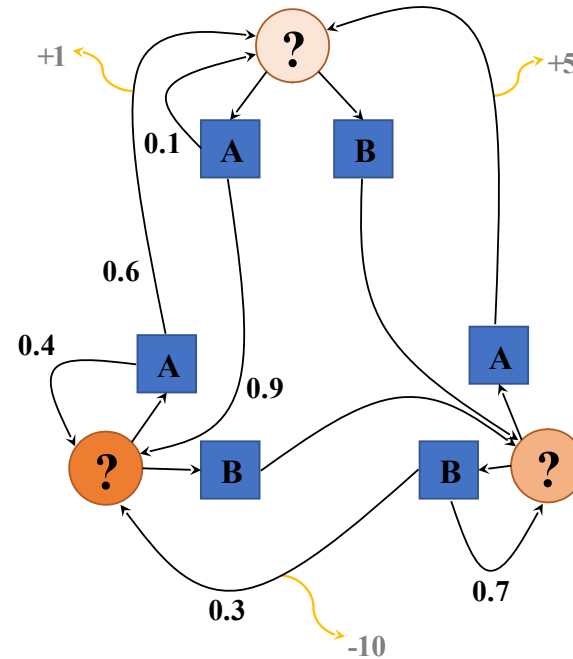
- $\mathcal{S}$ - State space
- $T : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$ - Transition probability distribution
- $\mathcal{A}$ - Action space
- $R : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ - Reward

Aleatory

Markov Decision Process (MDP)



Partially Observable Markov Decision Process (POMDP)

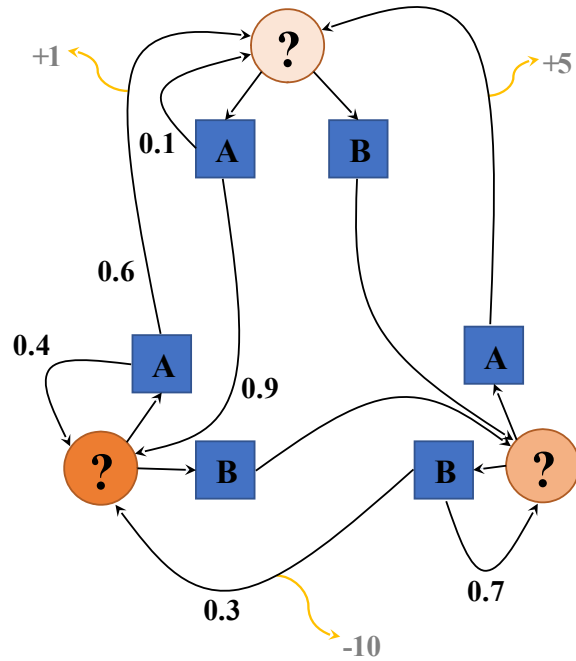


- $\mathcal{S}$ - State space
- $T : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$ - Transition probability distribution
- $\mathcal{A}$ - Action space
- $R : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ - Reward
- $\mathcal{O}$ - Observation space
- $Z : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \times \mathcal{O} \rightarrow \mathbb{R}$ - Observation probability distribution

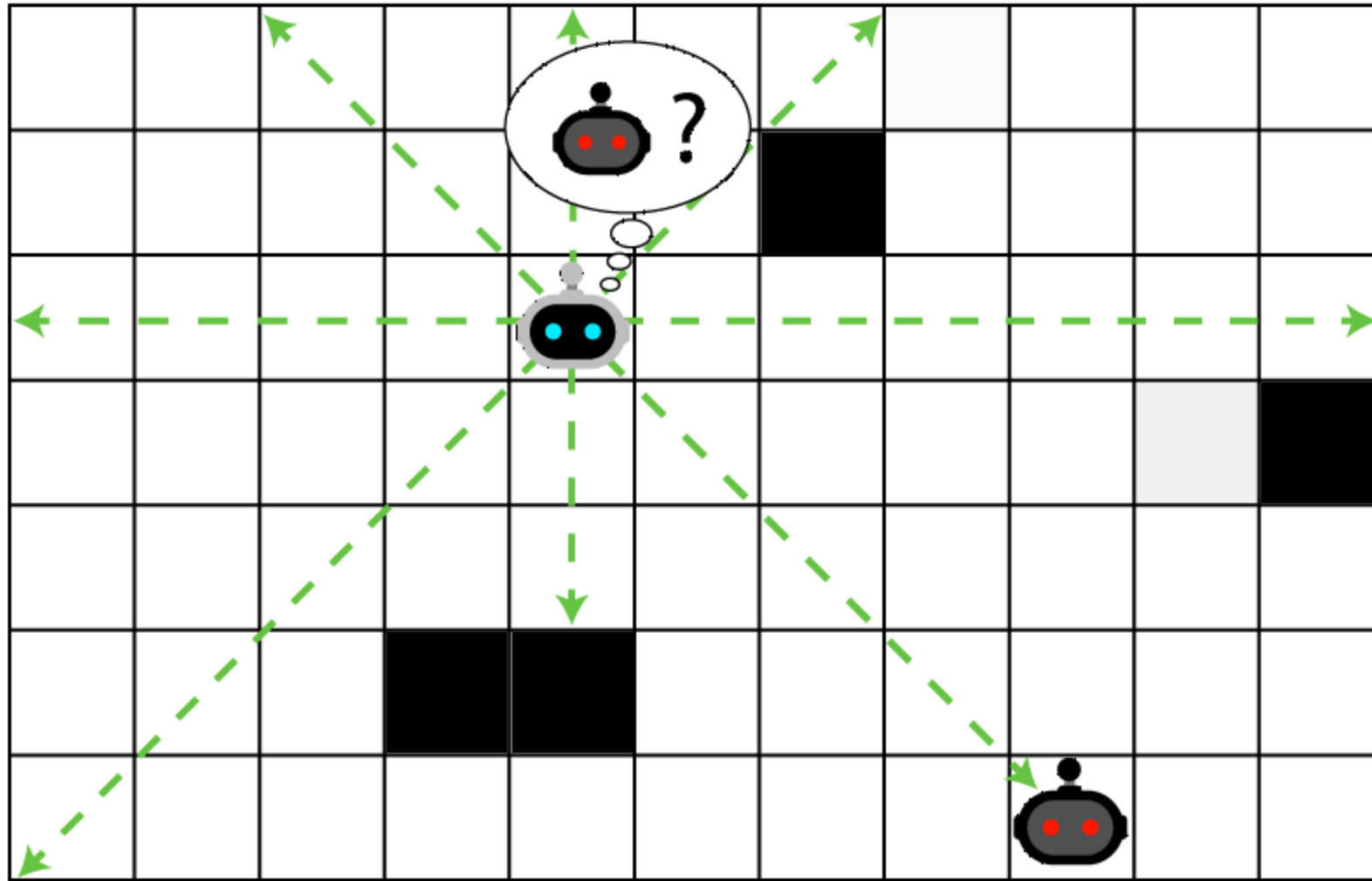
Aleatory

Epistemic

Partially Observable Markov Decision Process (POMDP)



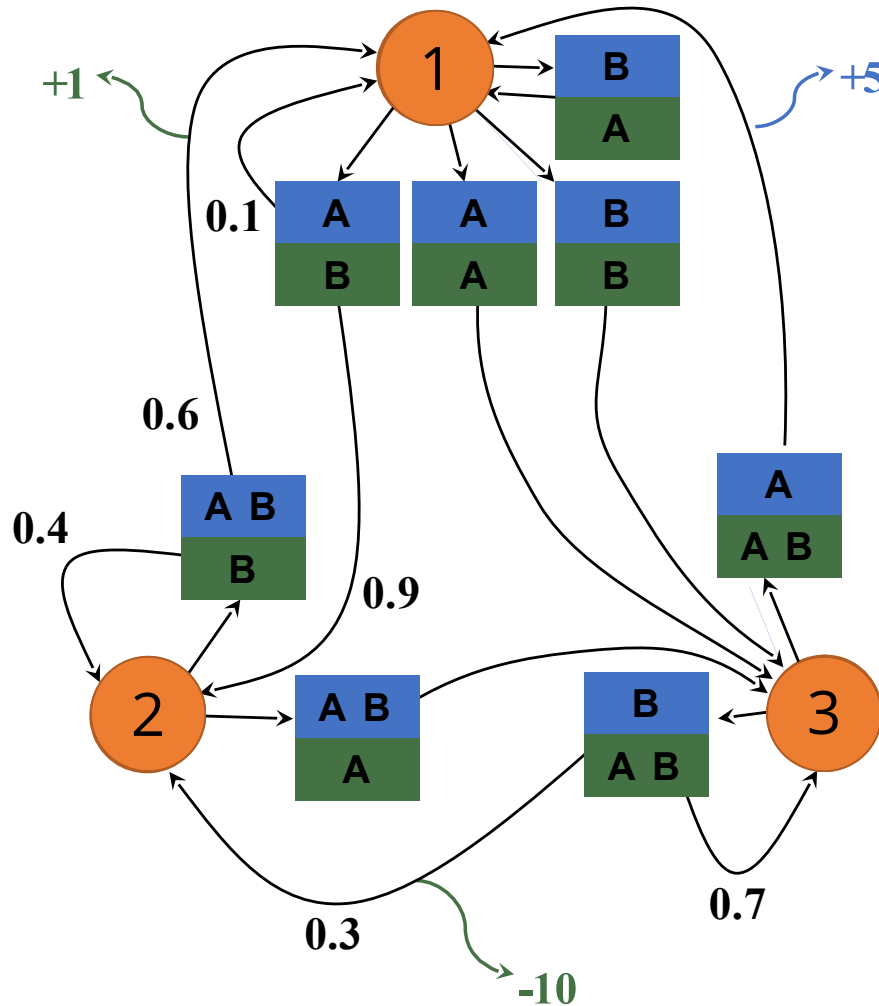
Laser Tag POMDP



Multi-agent environment posed as a single-agent problem

Evader policy predictably cowers in a corner

Markov Game (MG)

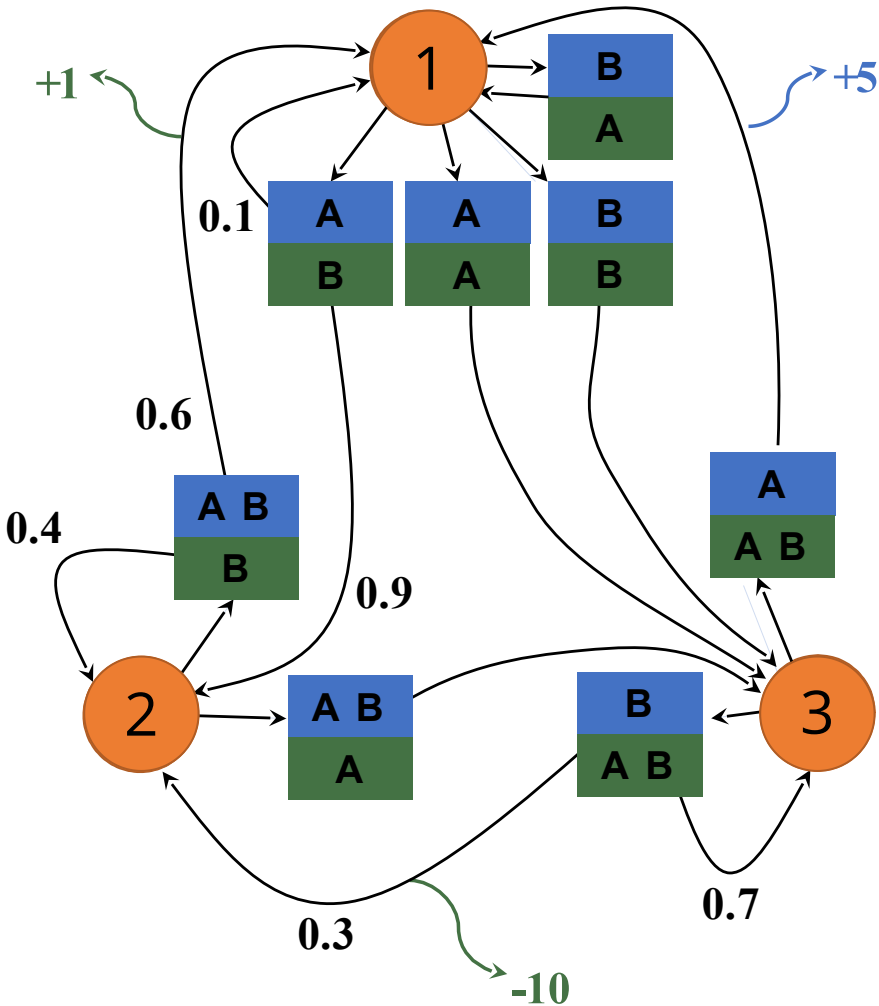


- $\mathcal{S}$ - State space
- $T(s' | s, \mathbf{a})$ - Transition probability distribution
- $\mathcal{A}^i, i \in 1..k$ - Action spaces
- $R^i(s, \mathbf{a})$ - Reward function (cooperative, opposing, or somewhere in between)

Aleatory

Interaction

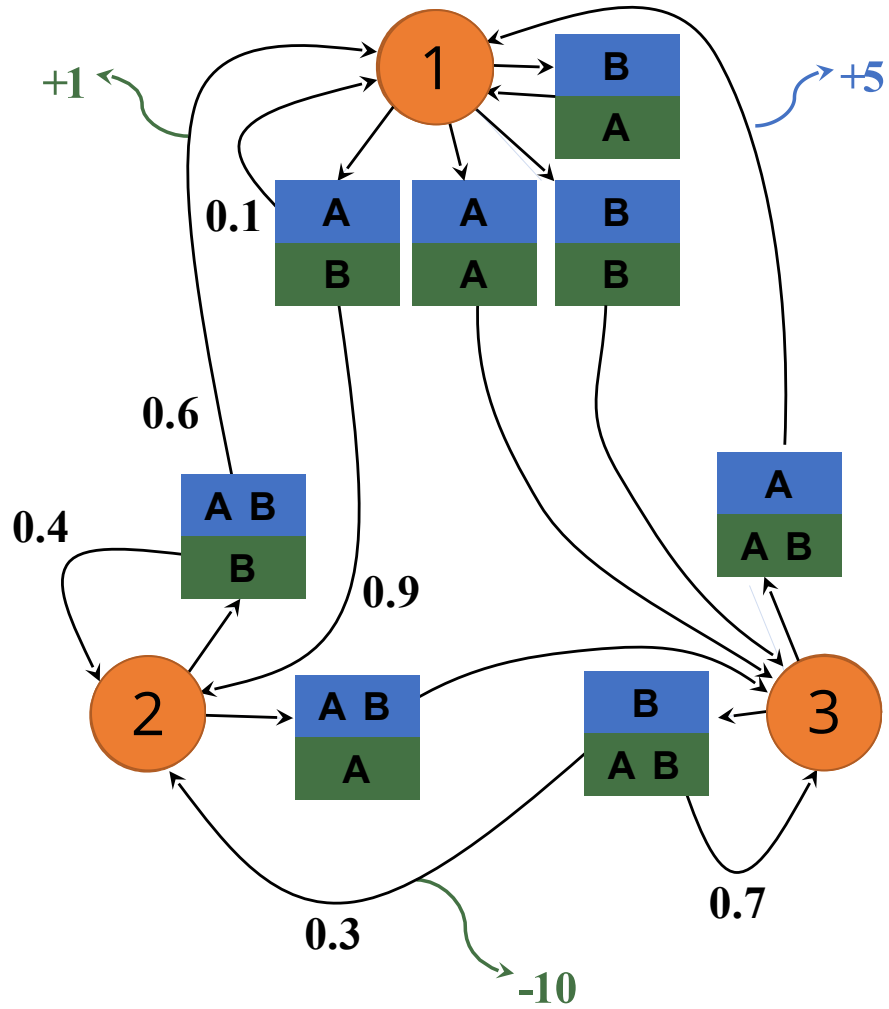
Markov Game (MG)



Nash Equilibrium

Incentive to deviate makes a policy suboptimal

Markov Game (MG)



The Prisoner's Dilemma

		Player 2	
		Cooperate	Defect
Player 1	Cooperate	<u>-1</u> , <u>-1</u>	<u>-10</u> , <u>0</u>
	Defect	<u>0</u> , <u>-10</u>	<u>-5</u> , <u>-5</u>

Two criminals are arrested for a crime. The police don't have enough evidence to convict them of a major crime, but they can convict them of a minor one. The police offer each prisoner a deal:

- If **one testifies (defects) against the other**, while the other remains silent (cooperates), the defector goes free, and the cooperator gets the full sentence (10 years).
- If **both defect**, they both get a medium sentence (5 years).
- If **both cooperate** (remain silent), they both get a short sentence (1 year).

The Prisoner's Dilemma

		Player 2	
		Cooperate	Defect
Player 1	Cooperate	-1, -1	-10, 0
	Defect	0, -10	-5, -5

The Prisoner's Dilemma

		Player 2	
		Cooperate	Defect
Player 1	Cooperate	-1, -1	-10, 0
	Defect	0, -10	-5, -5

The Prisoner's Dilemma

		Player 2	
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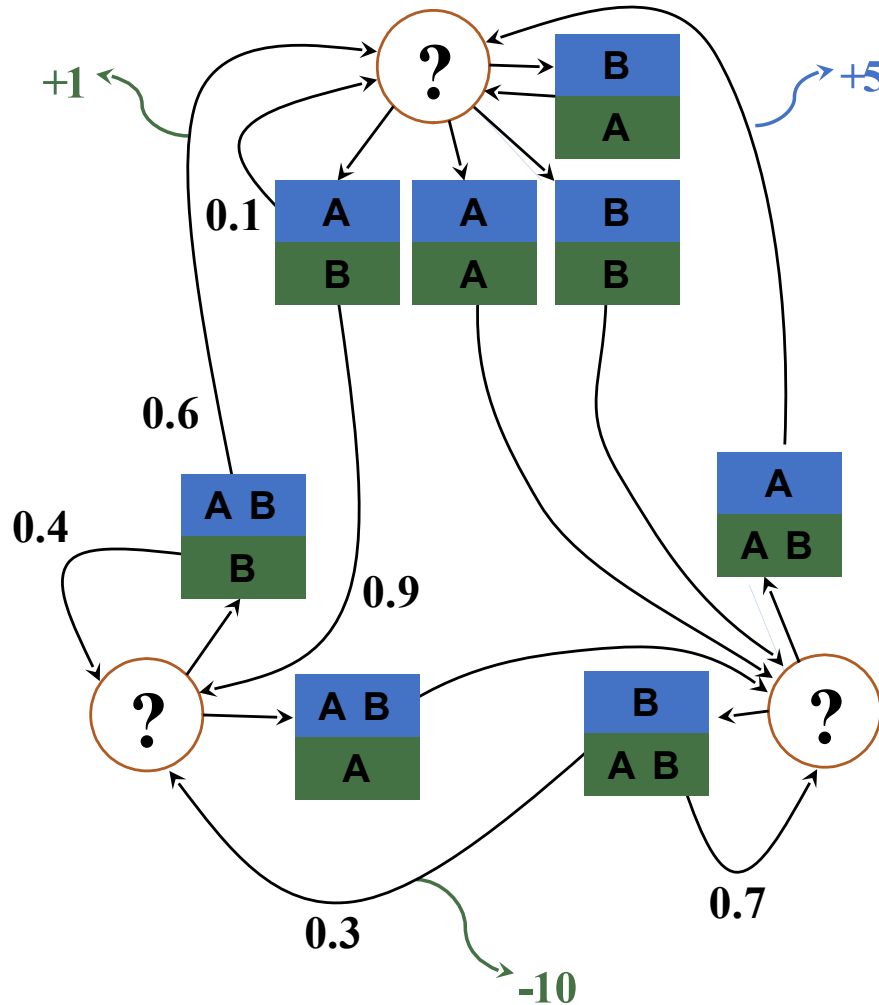
Two-player Zero-sum Games

Suppose I play Nash; opponent is somewhat irrational

Not only am I **guaranteed my Nash value**; opponent deviating from Nash can only increase my utility.

This is **not the case for general-sum games**. Irrational non-ego agents can bring you down with them with irrationality.

Partially Observable Stochastic Game (POSG)



- $\mathcal{S}$ - State space
- $T(s' | s, \mathbf{a})$ - Transition probability distribution
- $\mathcal{A}^i, i \in 1..k$ - Action spaces
- $R^i(s, \mathbf{a})$ - Reward function (cooperative, opposing, or somewhere in between)
- $\mathcal{O}^i, i \in 1..k$ - Observation spaces
- $Z(o^i | \mathbf{a}, s')$ - Observation probability distributions

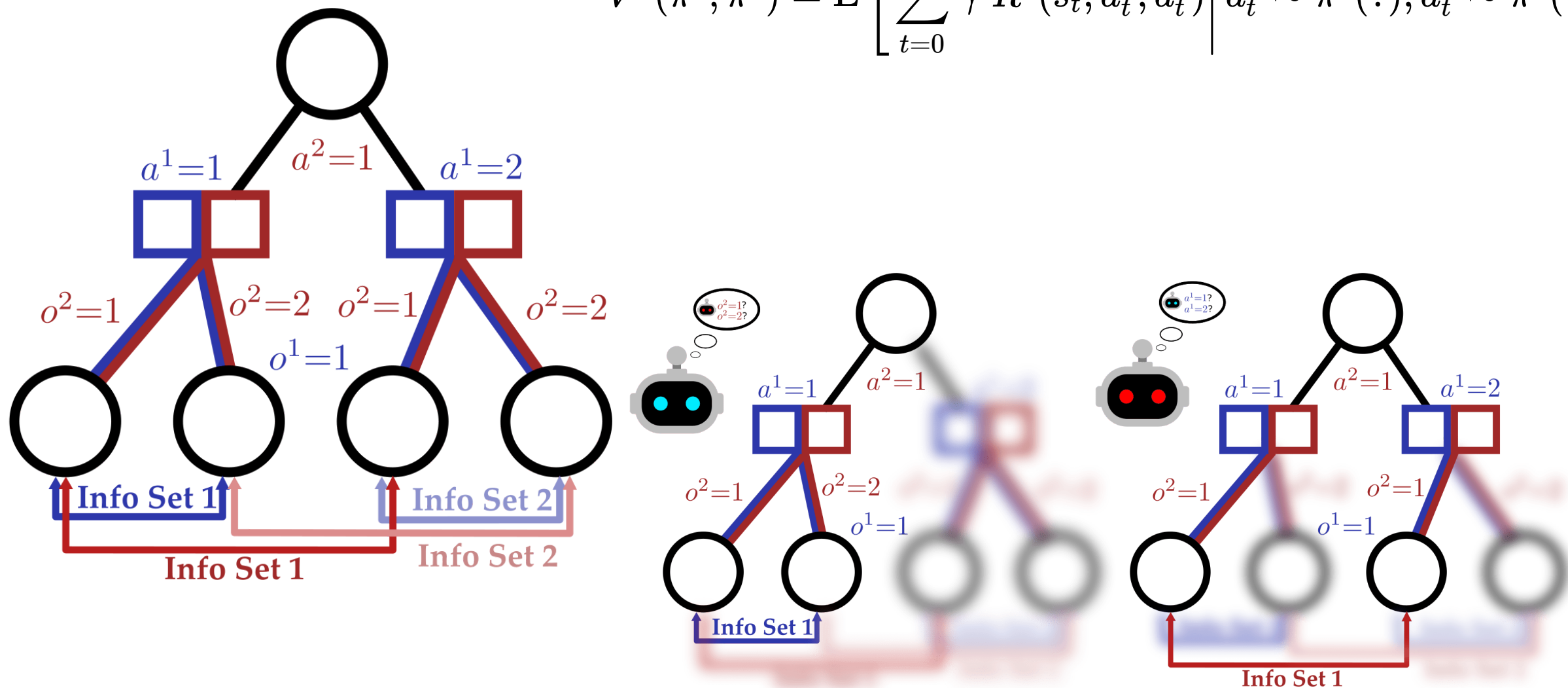
Aleatory

Epistemic

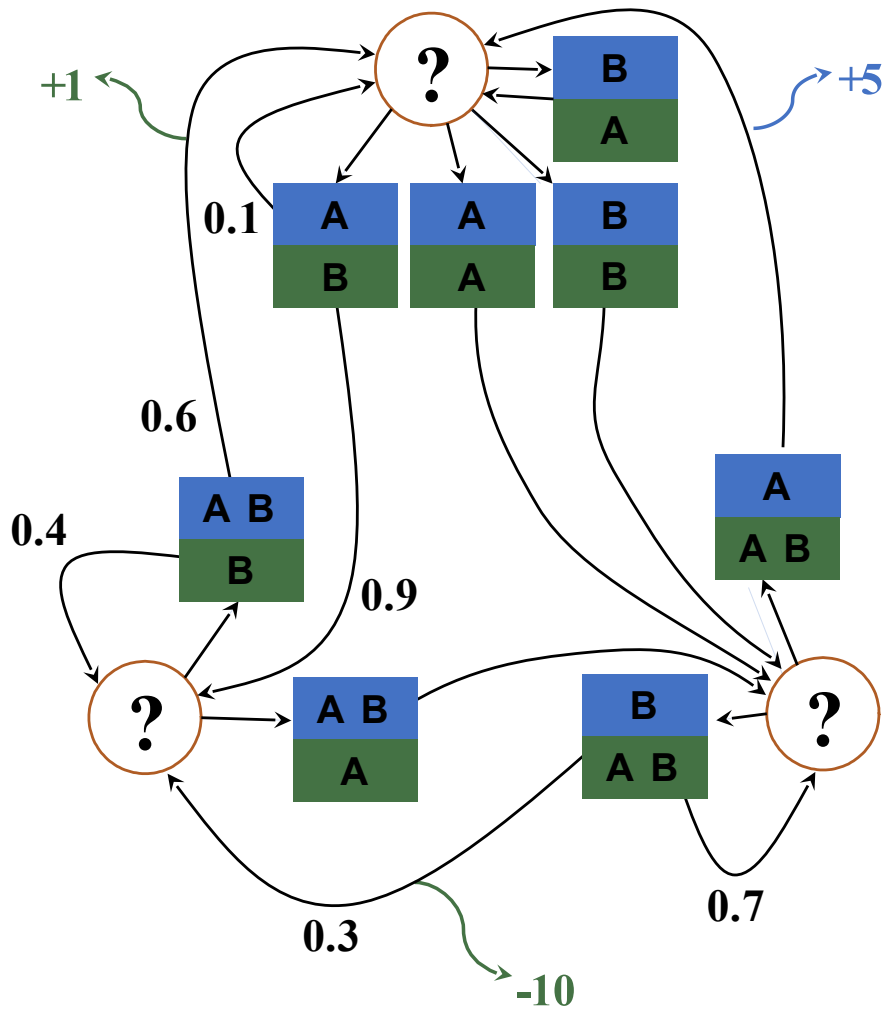
Interaction

Partially Observable Stochastic Game (POSG)

$$V^1(\pi^1, \pi^2) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R^1(s_t, a_t^1, a_t^2) \mid a_t^1 \sim \pi^1(?), a_t^2 \sim \pi^2(?) \right]$$



Partially Observable Stochastic Game (POSG)



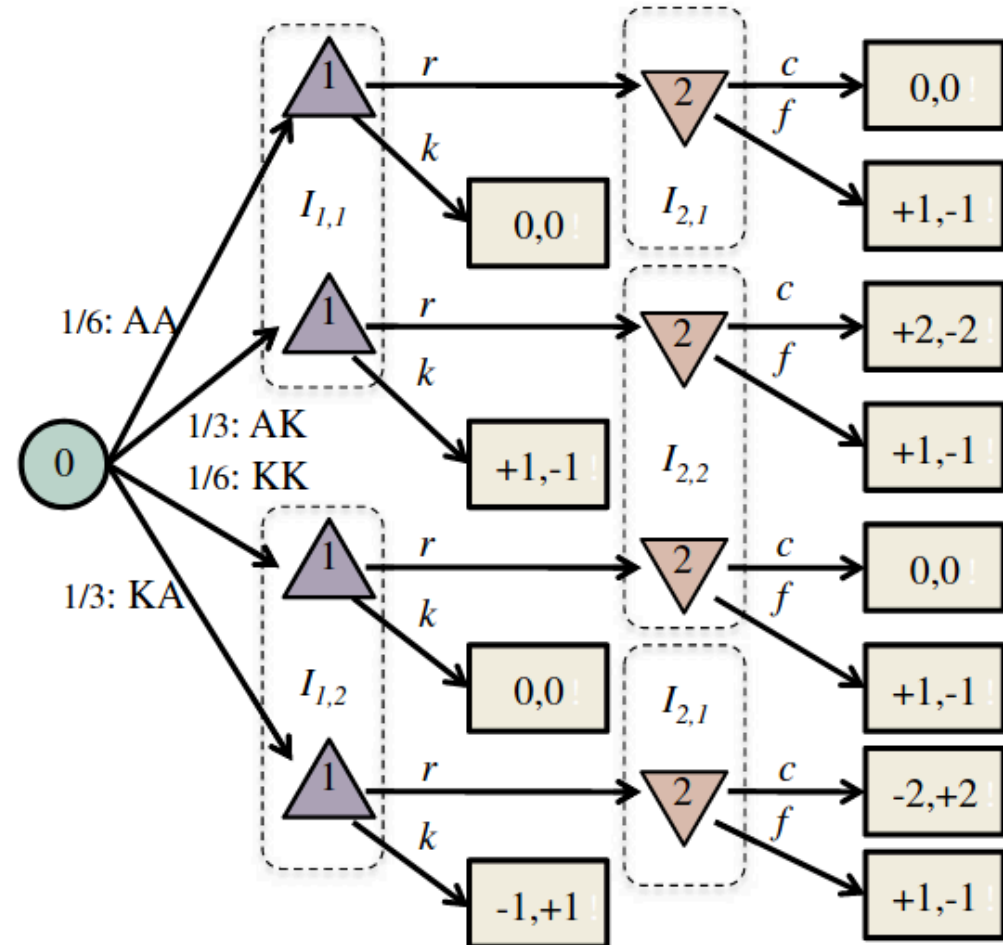
Extensive Form Game

Extensive-form game definition (h is a sequence of actions):

- Finite set of n players, plus the "chance" player
- $P(h)$ (player at each history)
- $A(h)$ (set of actions at each history)
- $I(h)$ (information set that each history maps to)
- $U(h)$ (payoff for each leaf node in the game tree)

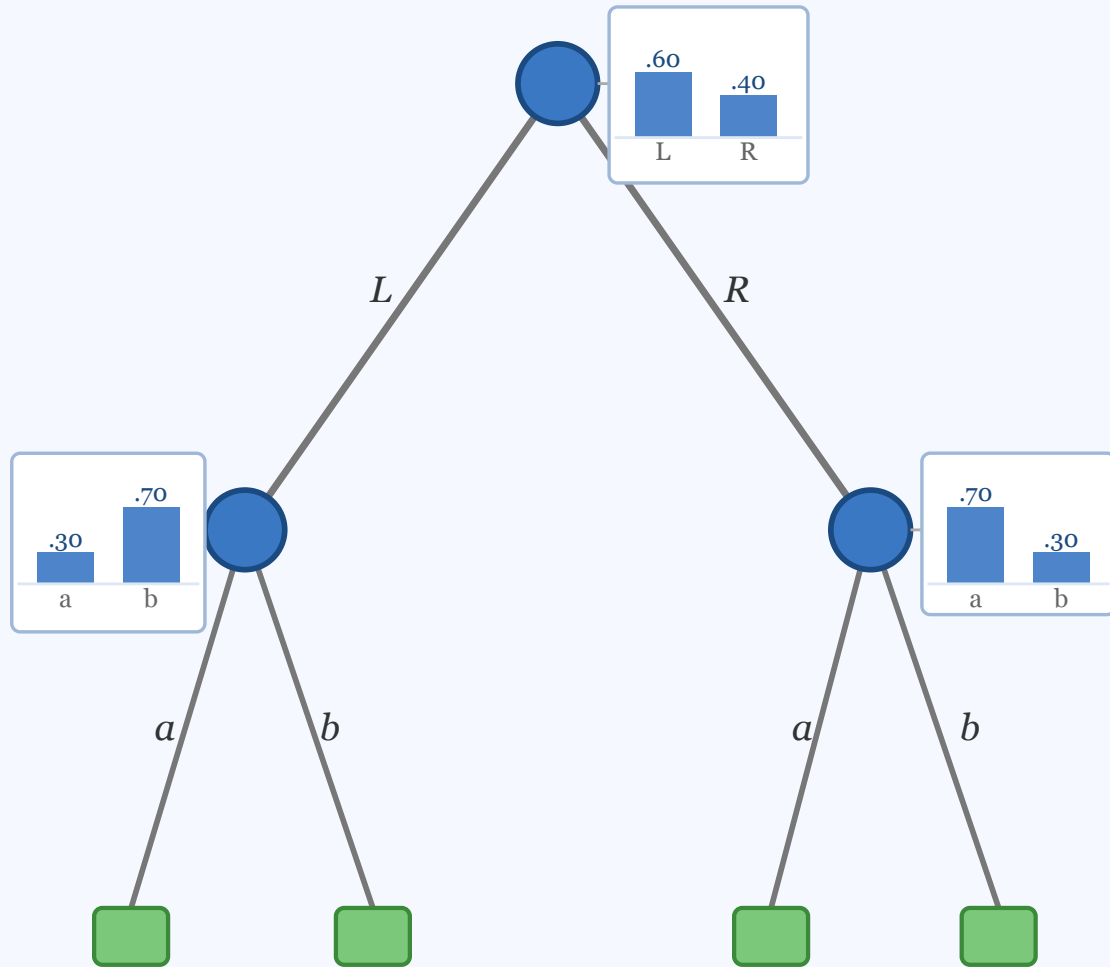
King-Ace Poker Example

- 4 Cards: 2 Aces, 2 Kings
- Each player is dealt a card
- P1 can either *raise* (r) the payoff to 2 points or *check* (k) the payoff at 1 point
- If P1 raises, P2 can either *call* (c) Player 1's bet, or *fold* (f) the payoff back to 1 point
- The highest card wins



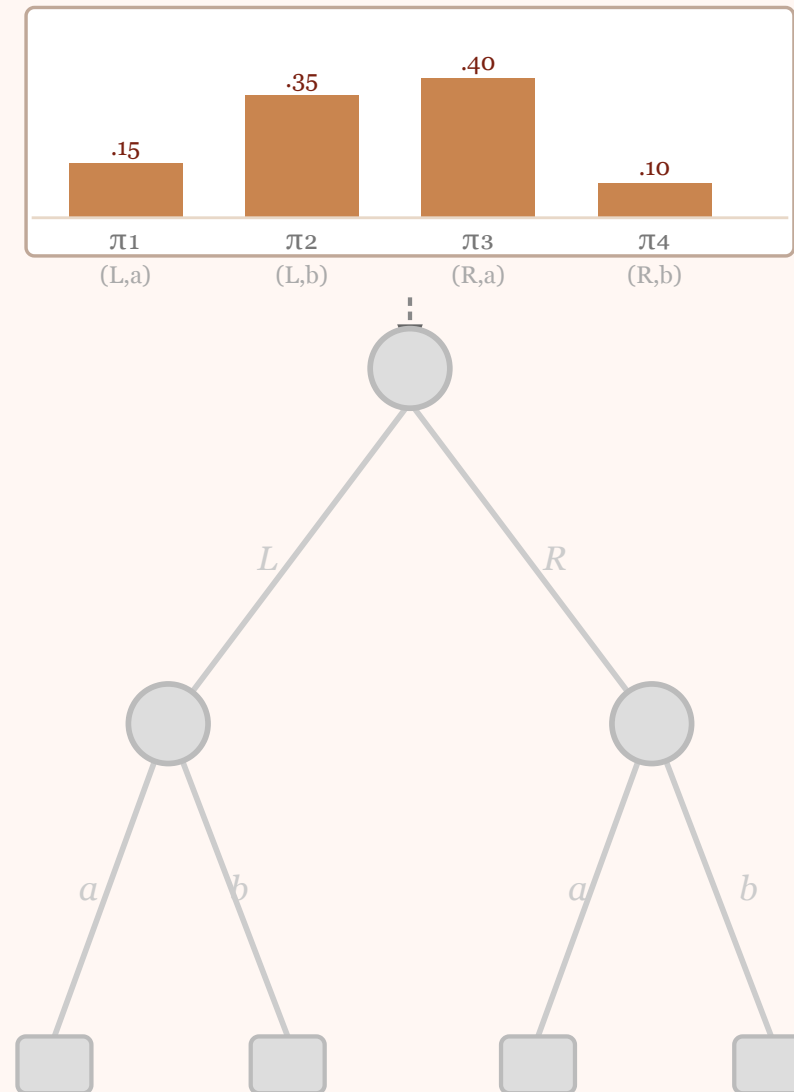
Behavioral Form

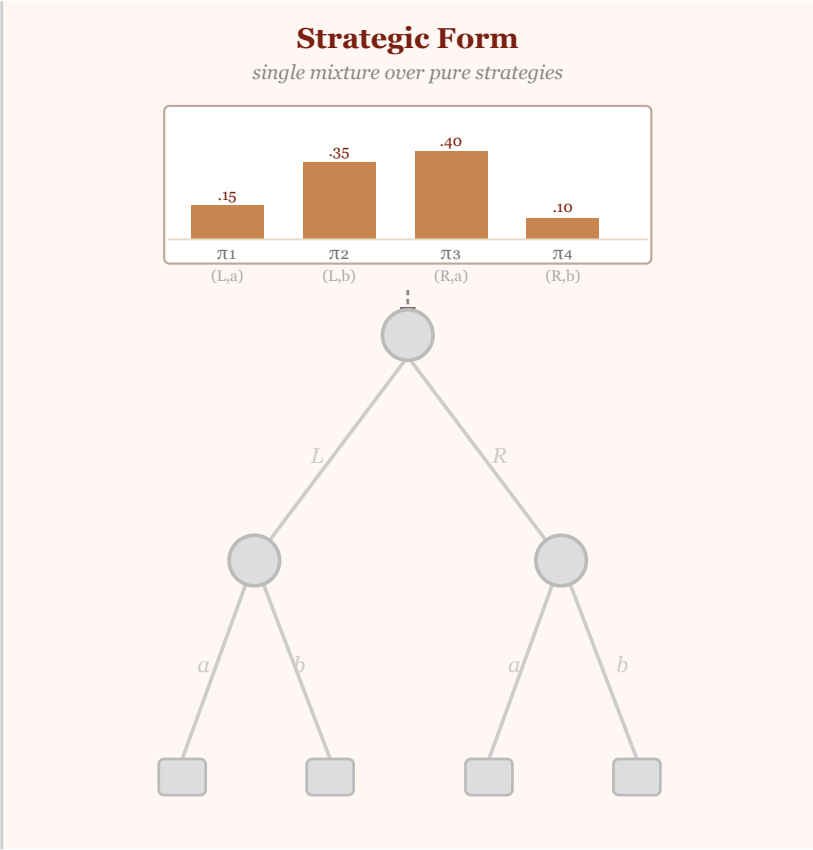
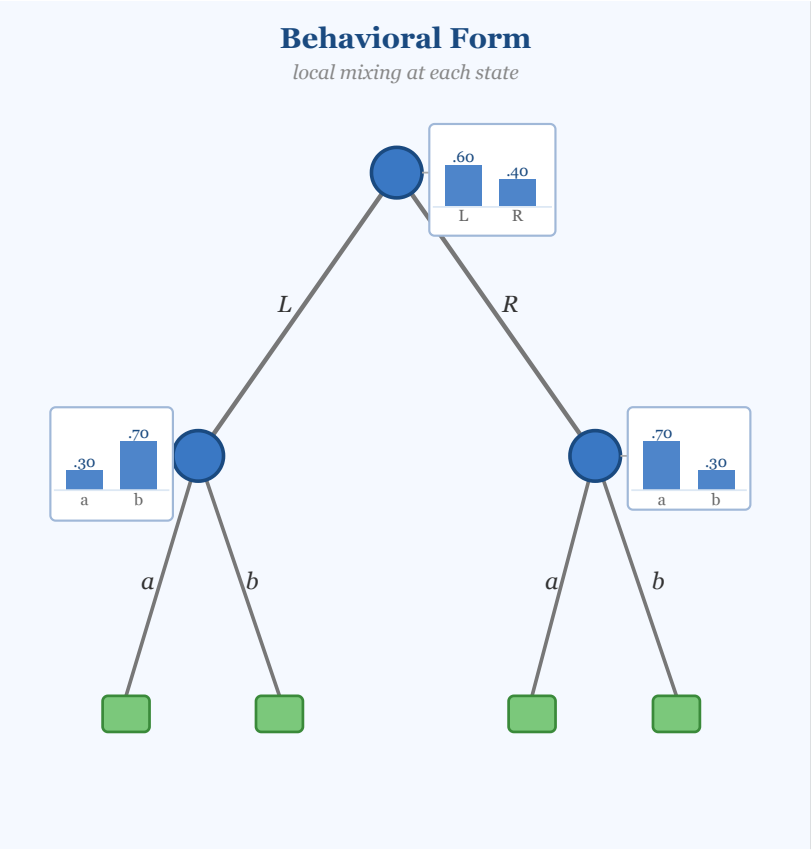
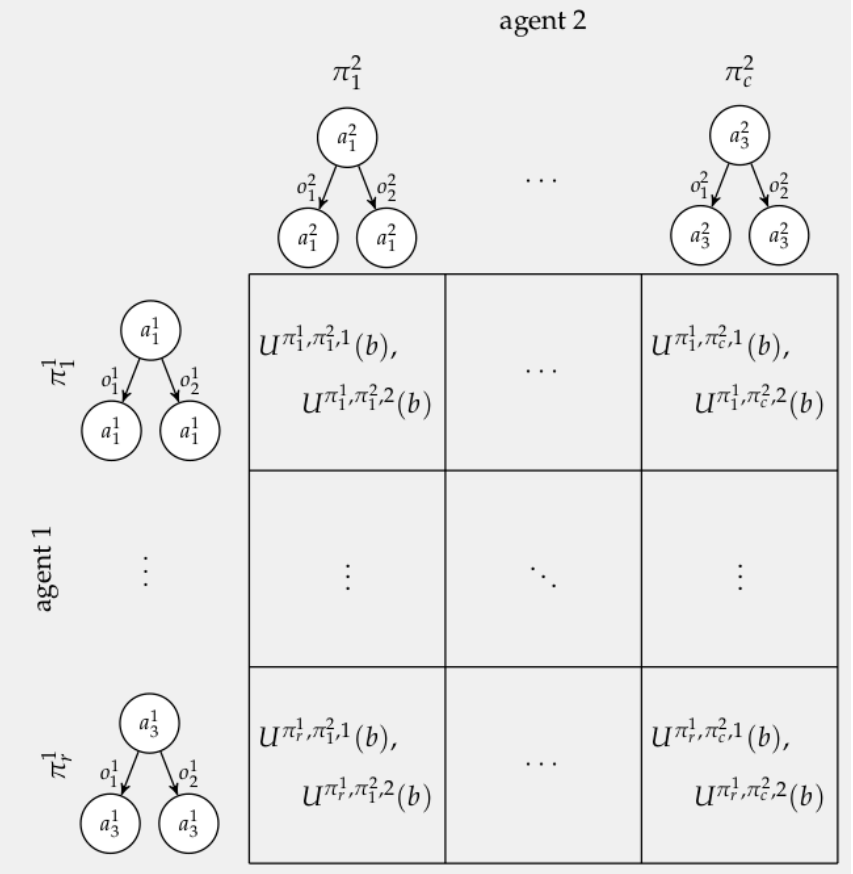
local mixing at each state



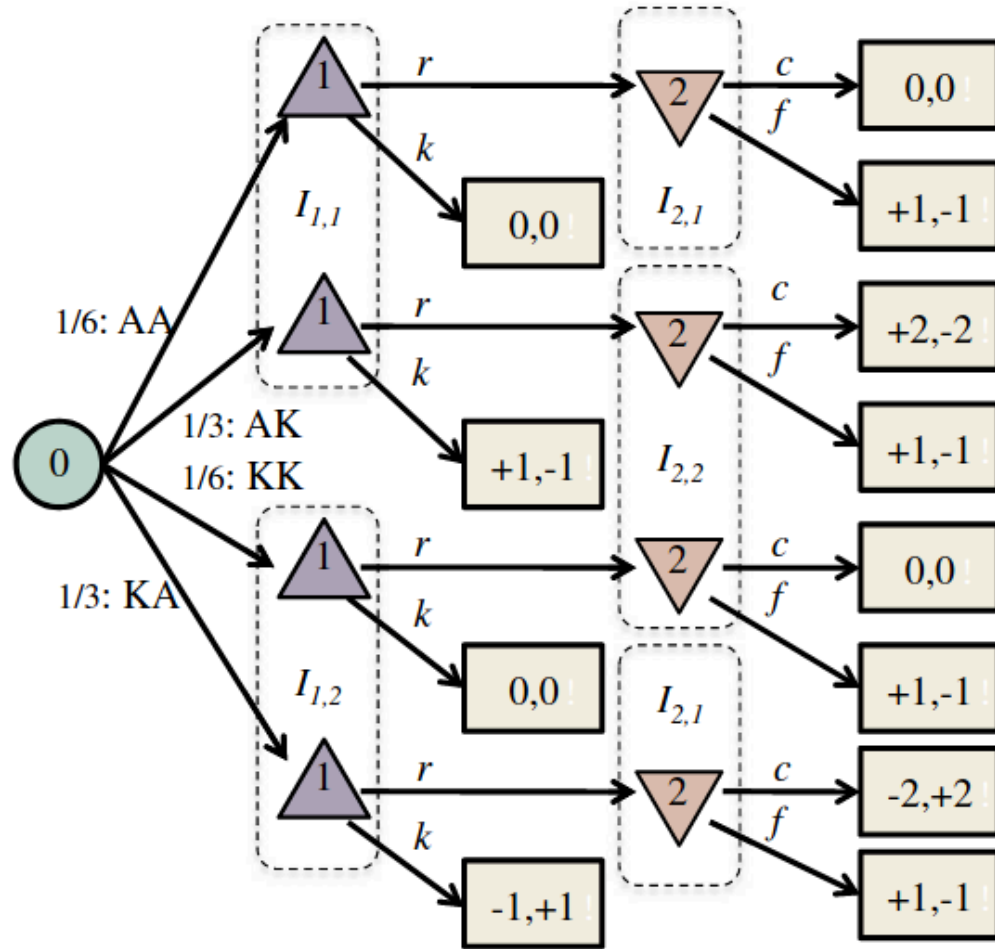
Strategic Form

single mixture over pure strategies





Extensive to Matrix Form



	2:cc	2:cf	2:ff	2:fc
1:rr	0	-1/6	1	7/6
1:kr	-1/3	-1/6	5/6	2/3
1:rk	1/3	0	1/6	1/2
1:kk	0	0	0	0

Fictitious Play in Extensive Form Games

Algorithm 2 General Fictitious Self-Play

function FICTITIOUSSELFPLAY(Γ, n, m)

Initialize completely mixed π_1

$\beta_2 \leftarrow \pi_1$

$j \leftarrow 2$

while within computational budget **do**

$\eta_j \leftarrow \text{MIXINGPARAMETER}(j)$

$\mathcal{D} \leftarrow \text{GENERATEDATA}(\pi_{j-1}, \beta_j, n, m, \eta_j)$

for each player $i \in \mathcal{N}$ **do**

$\mathcal{M}_{RL}^i \leftarrow \text{UPDATERLMEMORY}(\mathcal{M}_{RL}^i, \mathcal{D}^i)$

$\mathcal{M}_{SL}^i \leftarrow \text{UPDATESLMEMORY}(\mathcal{M}_{SL}^i, \mathcal{D}^i)$

$\beta_{j+1}^i \leftarrow \text{REINFORCEMENTLEARNING}(\mathcal{M}_{RL}^i)$

$\pi_j^i \leftarrow \text{SUPERVISEDLEARNING}(\mathcal{M}_{SL}^i)$

end for

$j \leftarrow j + 1$

end while

return π_{j-1}

end function

function GENERATEDATA(π, β, n, m, η)

$\sigma \leftarrow (1 - \eta)\pi + \eta\beta$

$\mathcal{D} \leftarrow n$ episodes $\{t_k\}_{1 \leq k \leq n}$, sampled from strategy profile σ

for each player $i \in \mathcal{N}$ **do**

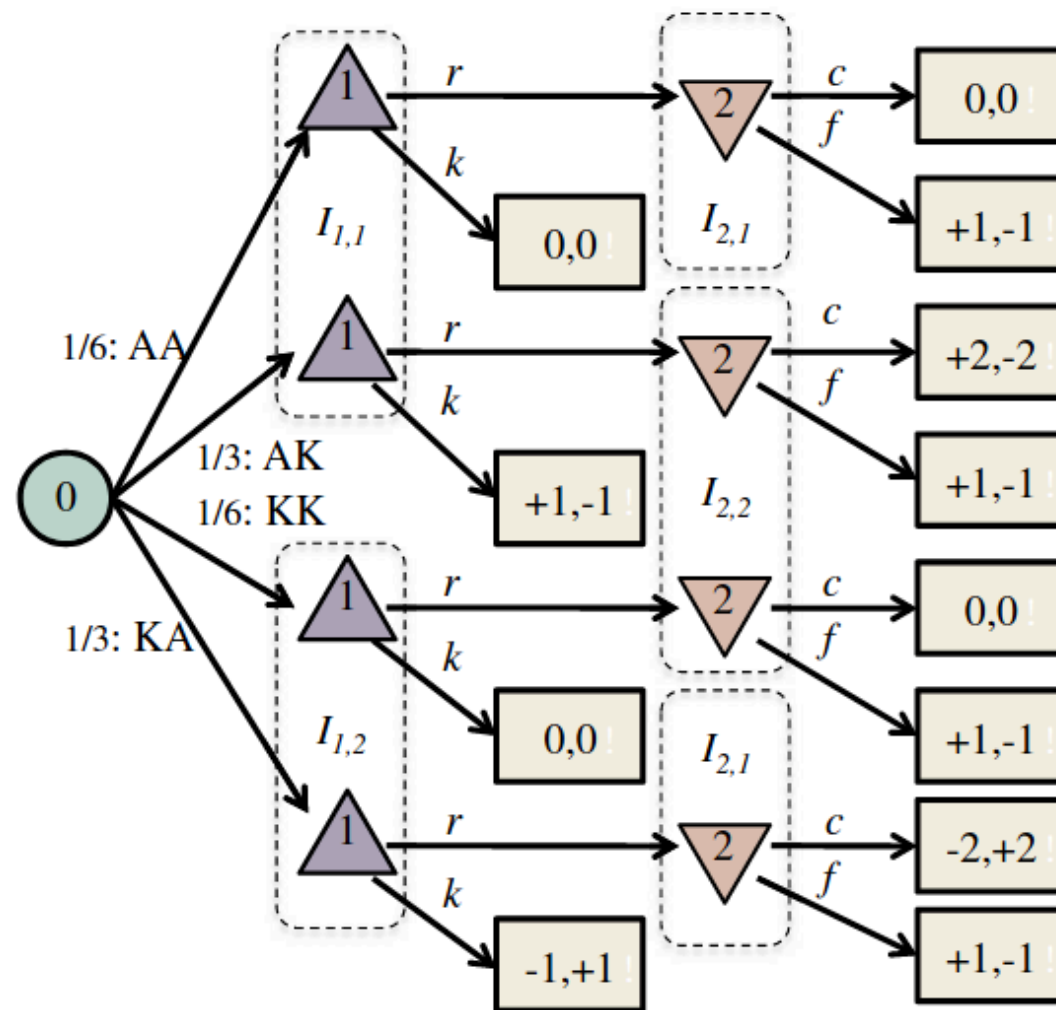
$\mathcal{D}^i \leftarrow m$ episodes $\{t_k^i\}_{1 \leq k \leq m}$, sampled from strategy profile (β^i, σ^{-i})

$\mathcal{D}^i \leftarrow \mathcal{D}^i \cup \mathcal{D}$

end for

return $\{\mathcal{D}^k\}_{1 \leq k \leq N}$

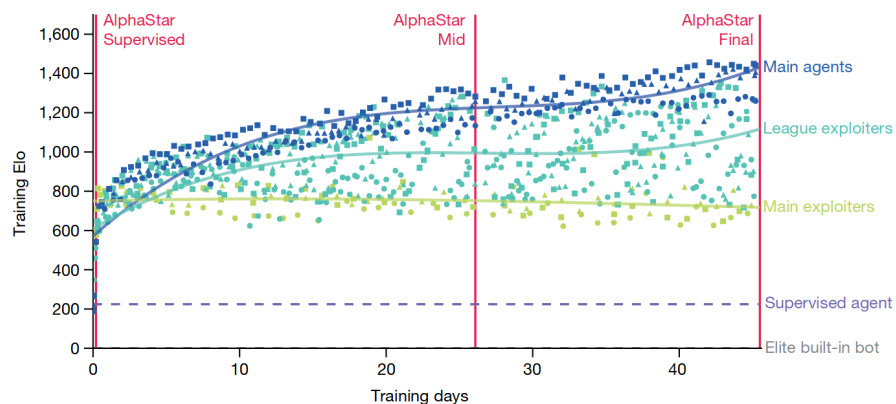
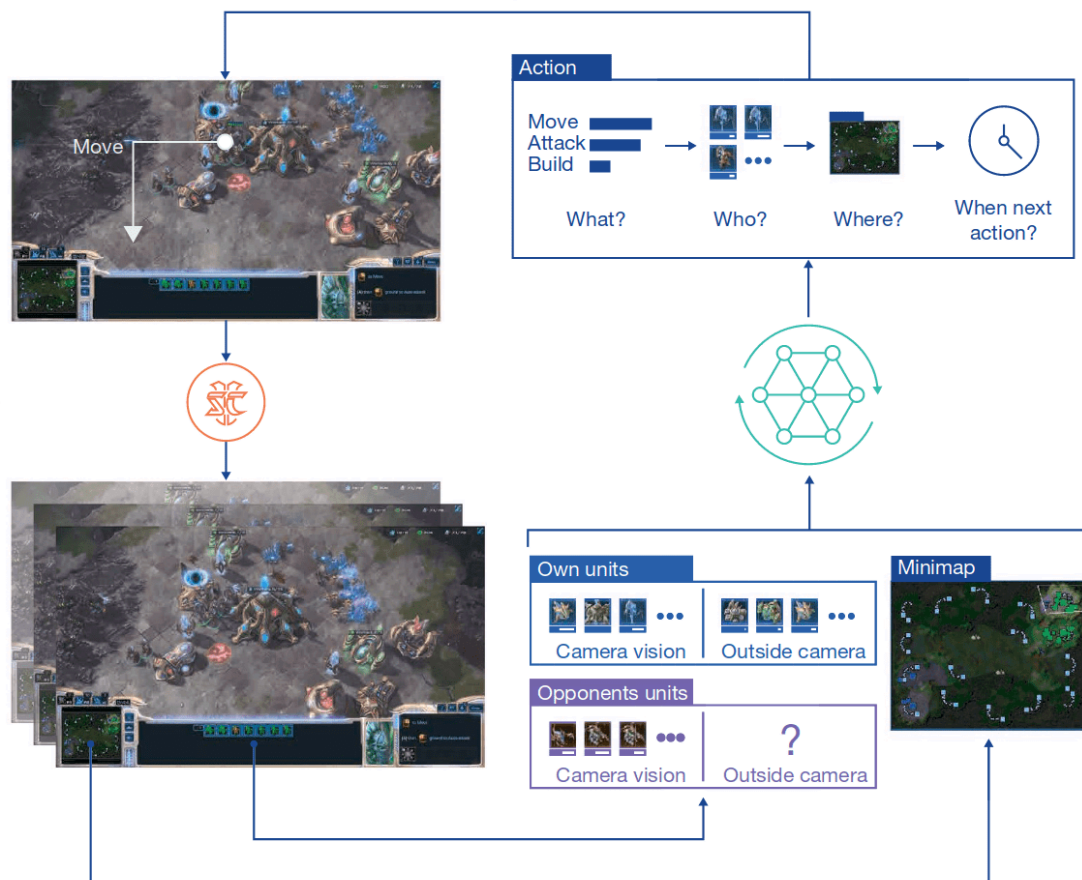
end function



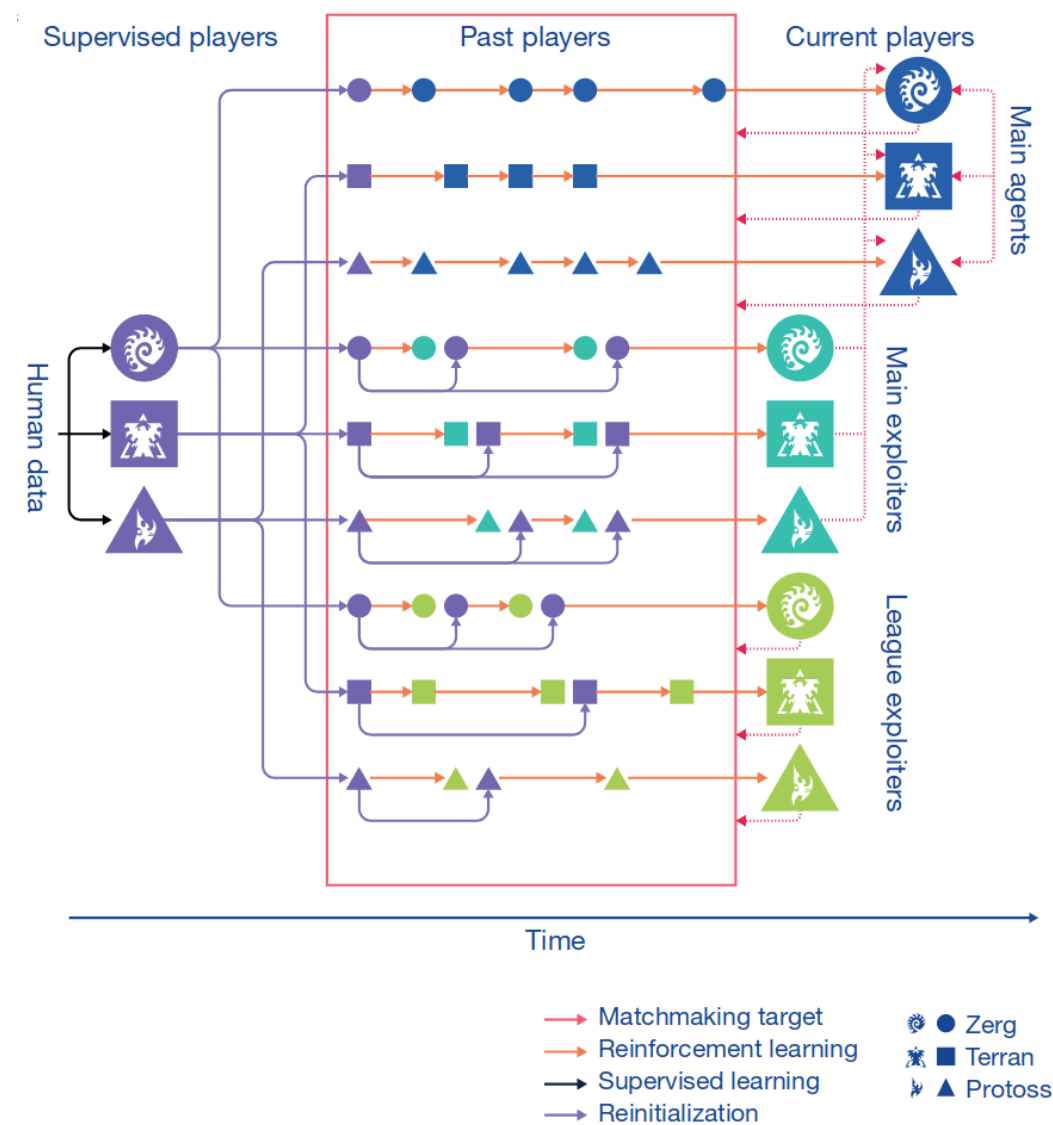
Actions limit ~22 per 5 s Requested delay ~200 ms

Real-time processing delay 80 ms

Real-time processing delay 30 ms



Alpha Star



Alpha Star

